

CS F407: Transformer Notes

Architecture notes

The original Transformer performs sequence-to-sequence translation. An encoder represents the input; a decoder generates the target while attending to that representation. The supplied notebook demonstrates encoder-decoder translation, decoder-only GPT-2 generation, and encoder-only sentiment analysis.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V.$$

Queries Q match keys K to weight values V ; mask M excludes positions. Self-attention obtains queries, keys, and values from one sequence. Cross-attention uses decoder queries and encoder keys/values. Multihead attention combines several learned projections. Positional information exposes token order. A causal mask blocks future tokens during autoregressive generation.

Encoder only. Bidirectional context supports classification, retrieval, and representations. **Decoder only.** Causal context supports next-token generation. **Encoder-decoder.** Input encoding and output generation support translation and other sequence-to-sequence tasks.

TODO

Supplied exercise	Status
BERT-to-BERT English/German translation	TODO
T5 English/French translation	TODO
English/Hinglish translation	TODO
GPT-2 generation	TODO
Sentiment pipeline	TODO
Ollama Mistral question answering	TODO

These exercises require pretrained model downloads. Ollama also requires a running server and Mistral weights. Its terminal cells use Colab extensions, so local execution needs an appropriate terminal/server setup. A comparison with another assistant should use the same prompts and record actual responses. Fluency does not establish factual correctness.